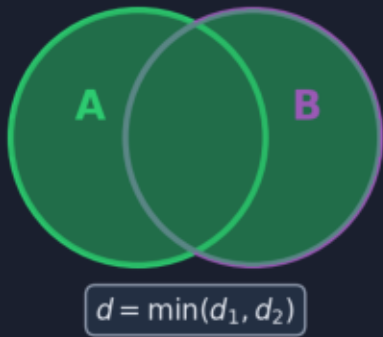


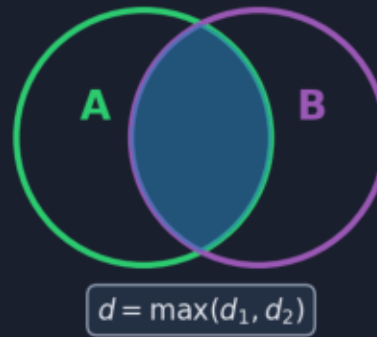
01 boolean venn

Boolesche Operationen → SDF-Kombination

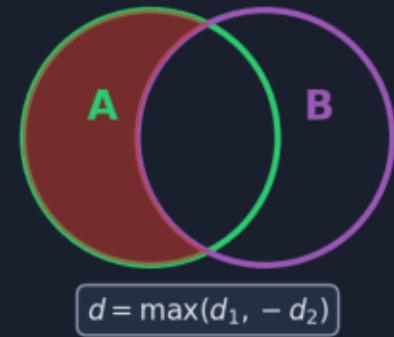
Vereinigung $A \cup B$



Schnittmenge $A \cap B$

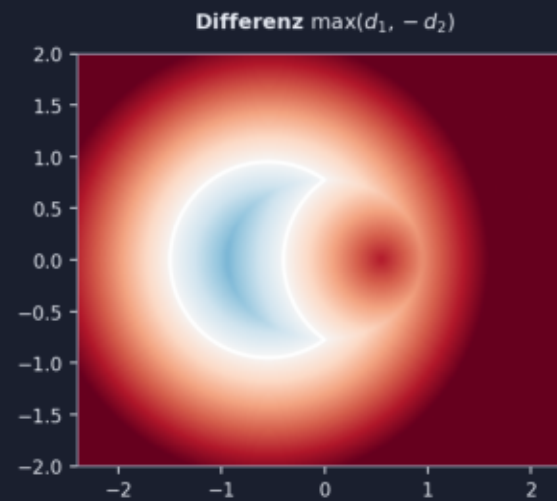
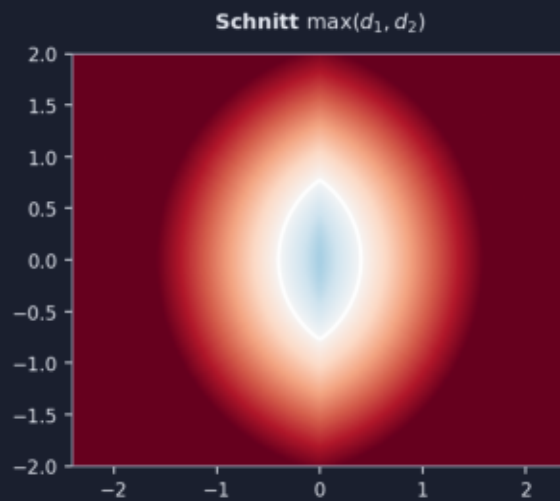
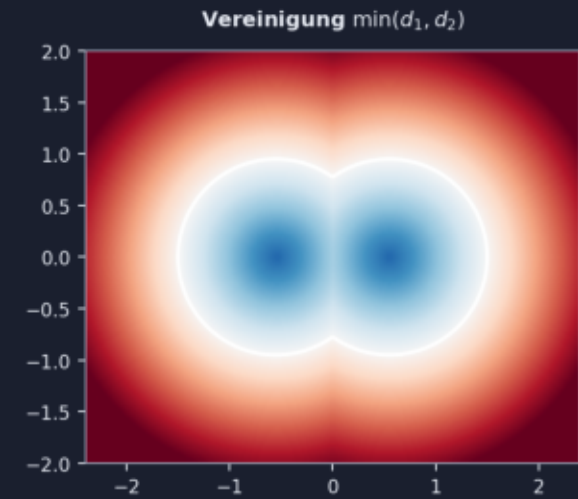
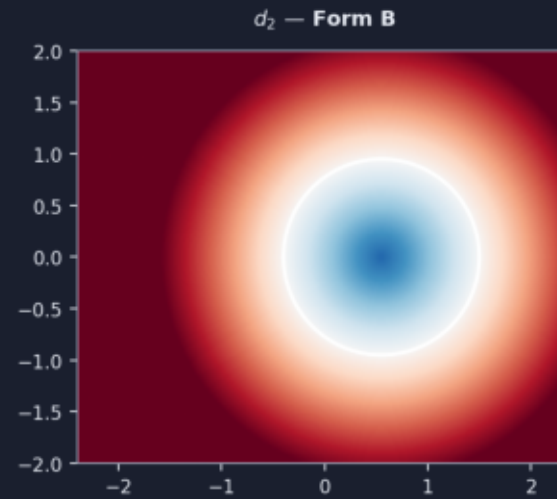
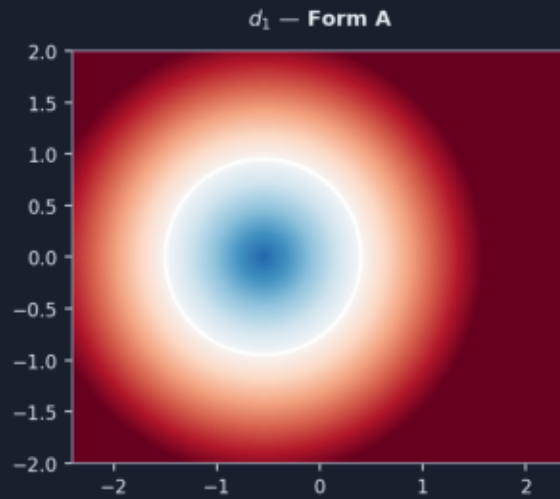


Differenz $A \setminus B$

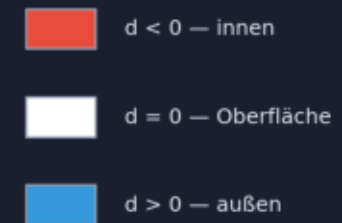


02 sdf heatmap panel

2D-Querschnitt: Distanzfelder vor und nach CSG



Signed Distance Field

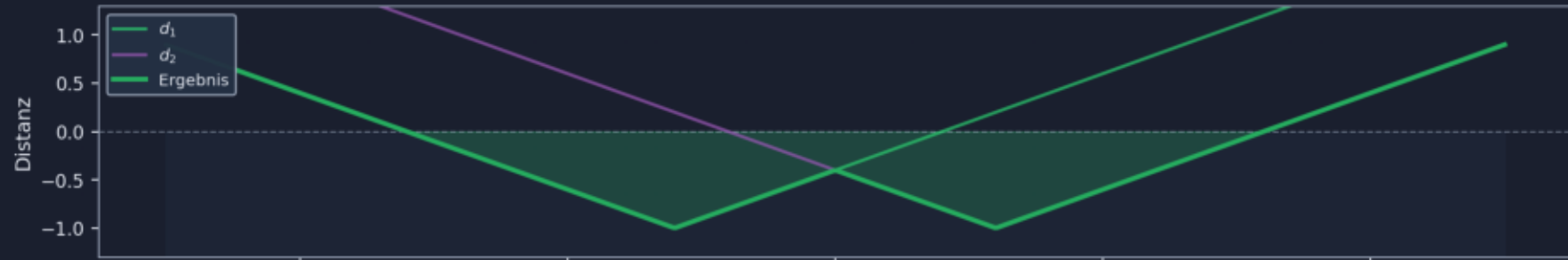


Rot = nah an/vor der Fläche (innen)
Blau = weit draußen

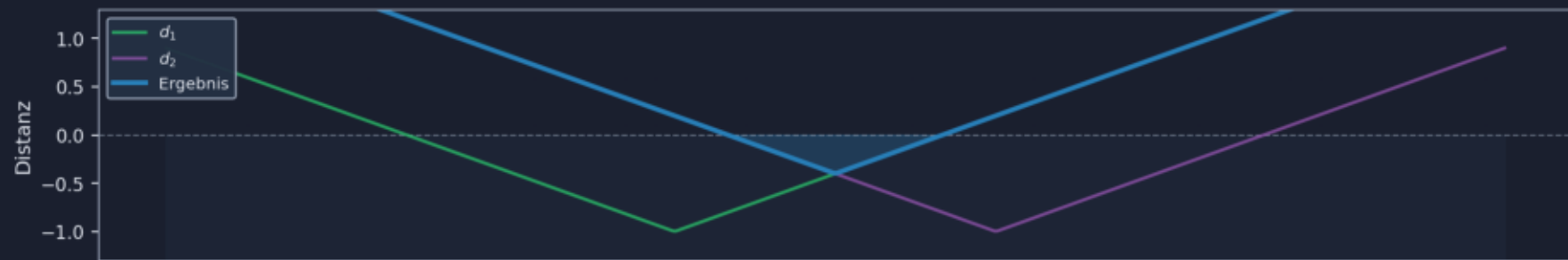
03 1d scanline

1D-Intuition: horizontale Probe durch zwei Kreise

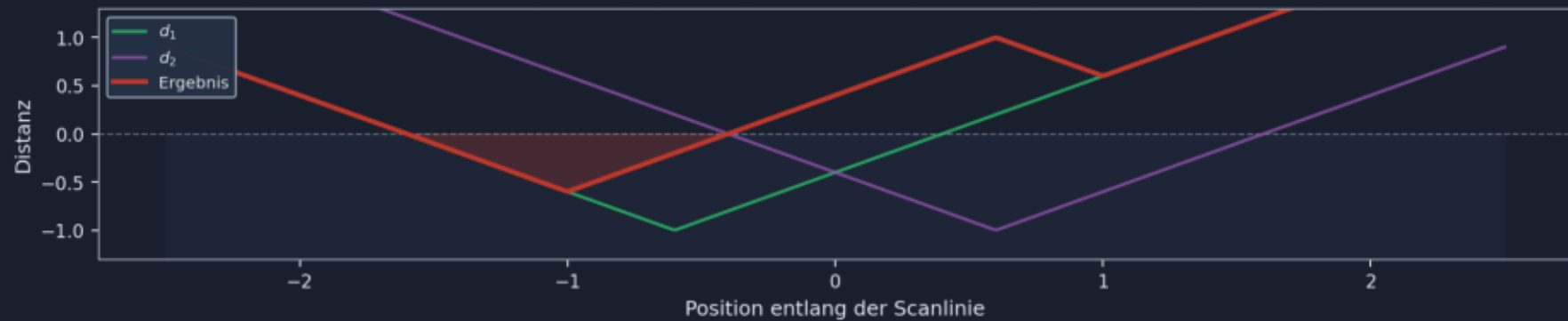
Vereinigung: $\min(d_1, d_2)$



Schnitt: $\max(d_1, d_2)$



Differenz: $\max(d_1, -d_2)$



04 inside outside table

Warum min / max? — Innen iff alle relevanten $d_i < 0$

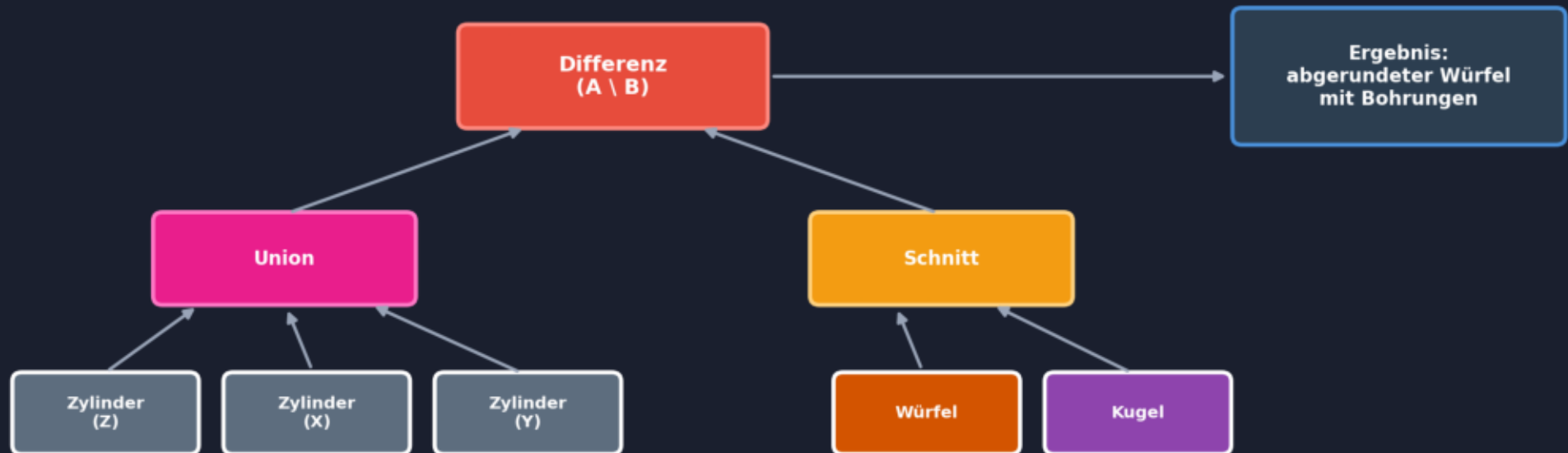
Punktlage	d_1	d_2	Union min	Schnitt max	Diff max($\cdot, -$)
nur in A	< 0	> 0	$< 0 \checkmark$	$> 0 \times$	$< 0 \checkmark$
nur in B	> 0	< 0	$< 0 \checkmark$	$> 0 \times$	$> 0 \times$
in $A \cap B$	< 0	< 0	$< 0 \checkmark$	$< 0 \checkmark$	$< 0 \checkmark$
außen	> 0	> 0	$> 0 \times$	$> 0 \times$	$> 0 \times$

05 csg tree

CSG-Baum: hierarchische Konstruktion

Blätter = Primitive SDFs · Knoten = boolesche Operatoren · Wurzel = finale Szene-SDF

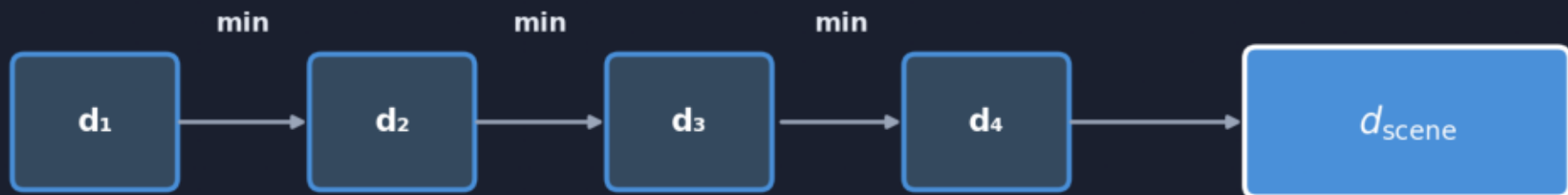
```
float scene(vec3 p) {  
    float holes = opUnion(opUnion(cylZ(p), cylX(p)), cylY(p));  
    float core = opIntersect(box(p), sphere(p));  
    return opSubtract(core, holes);  
}
```



06 nary union chain

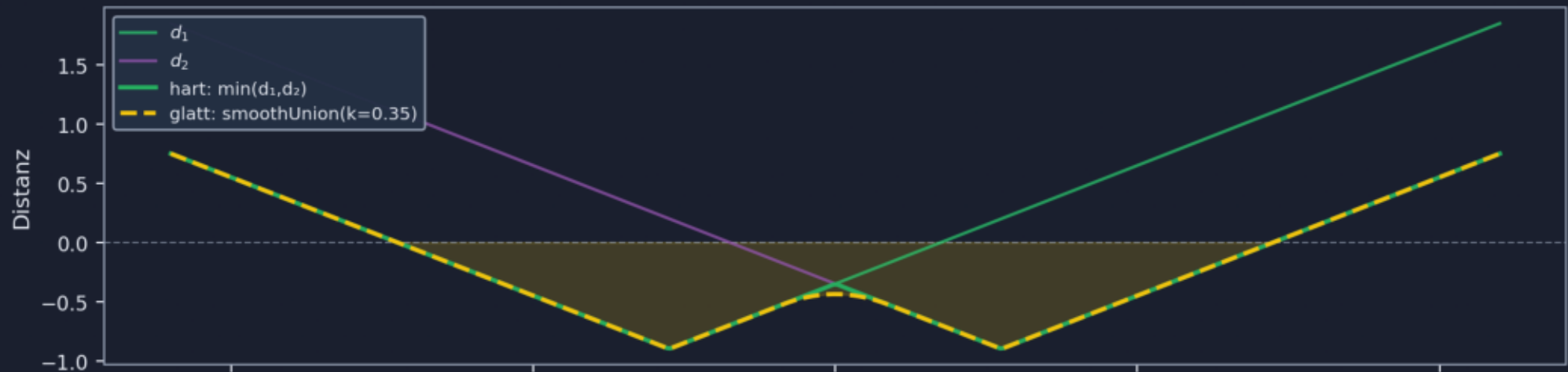
N-äre Vereinigung = wiederholtes min

```
float d = d1;  
d = min(d, d2);  
d = min(d, d3);  
d = min(d, d4);
```

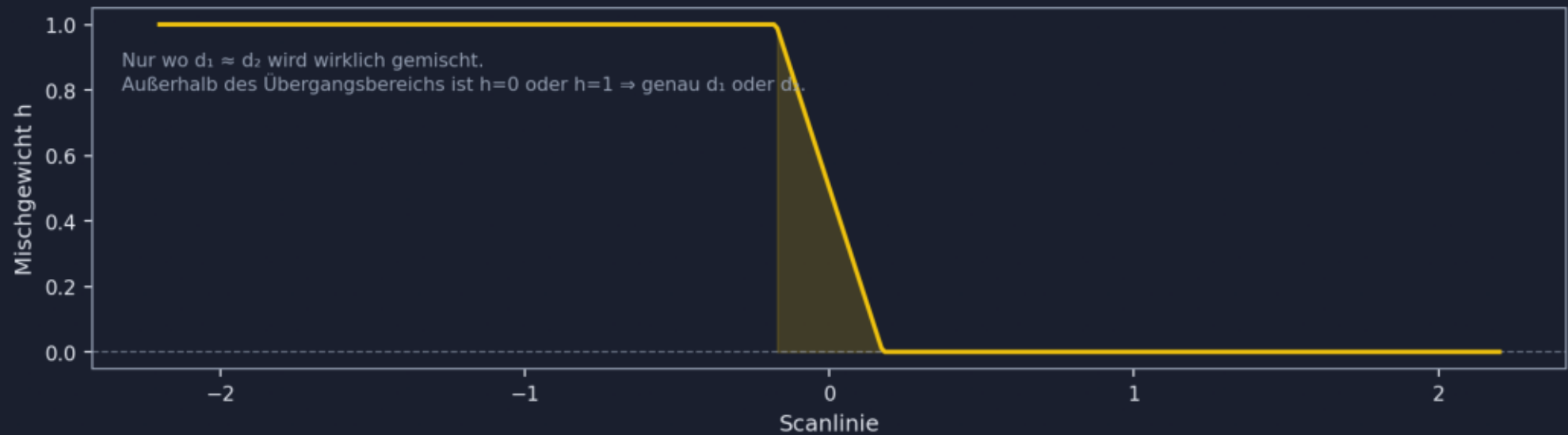


07 smooth union

Glatte CSG-Union: polynomiale Mischung statt hartem min/max
Harte vs. glatte Union entlang einer Scanlinie



Mischfaktor $h(d_1, d_2) \in [0, 1]$



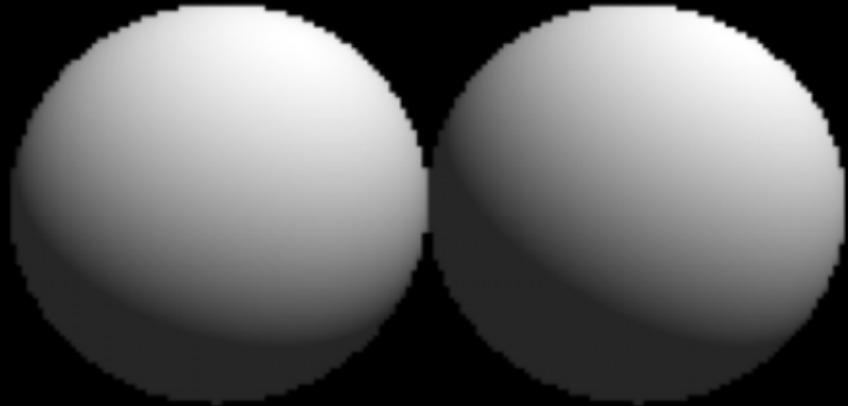
08 render difference



08 render intersection



08 render union



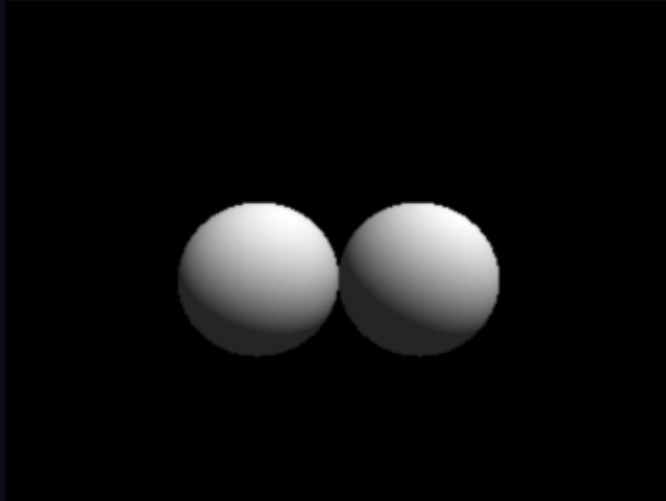
09 render csg tree



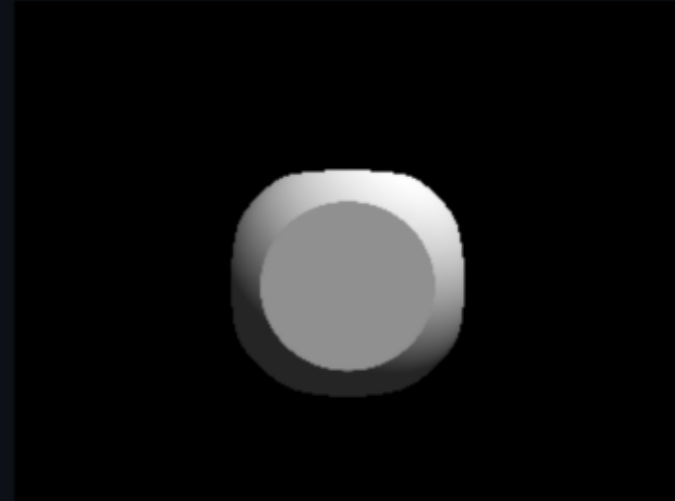
10 render panel

Ray-Marched CSG-Beispiele (aus SDF-Formeln)

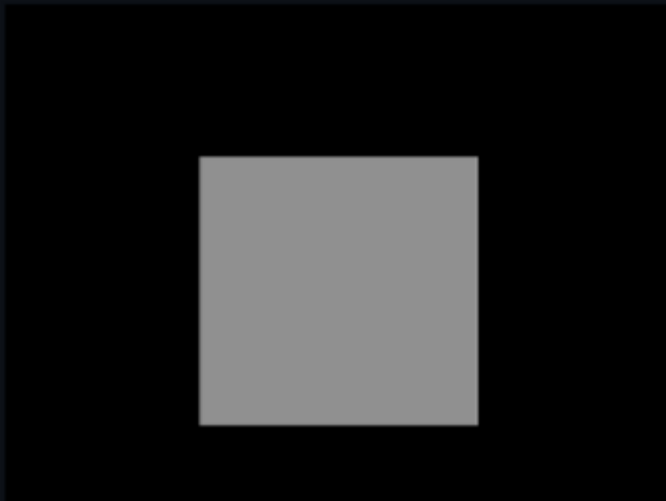
Union zweier Kugeln



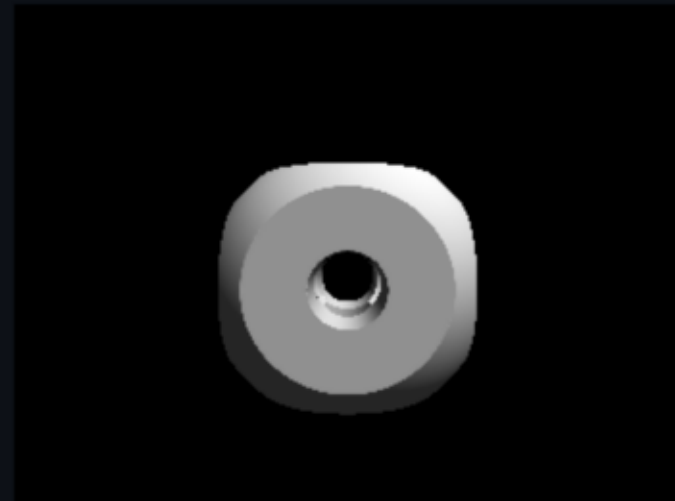
Schnitt Box $\cap$ Kugel



Differenz Box $\setminus$ Kugel



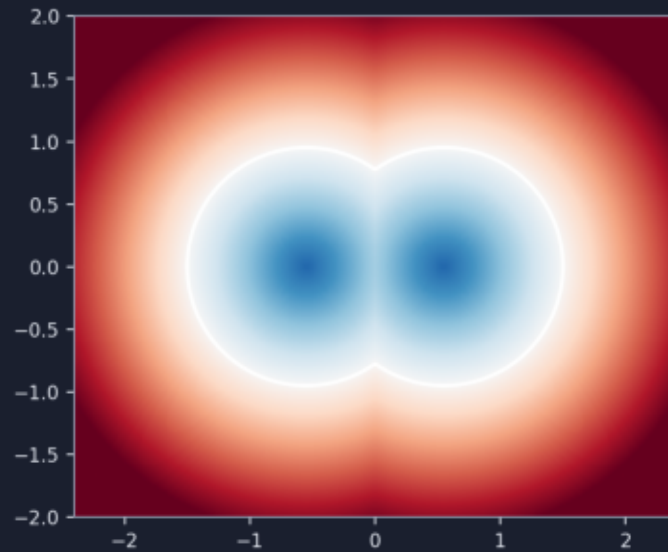
CSG-Baum: Würfel mit Bohrungen



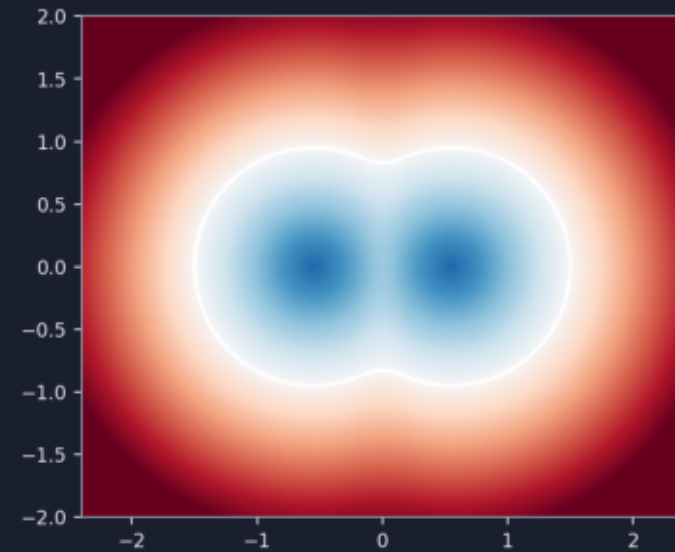
11 smooth 2d comparison

Wie k den Blend-Bereich steuert: klein = fast hart, groß = breite Verschmelzung

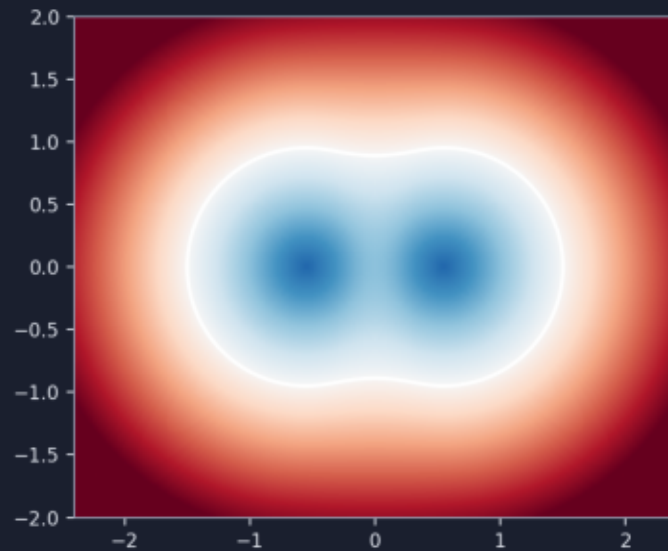
Harte Union: $\min(d_1, d_2)$



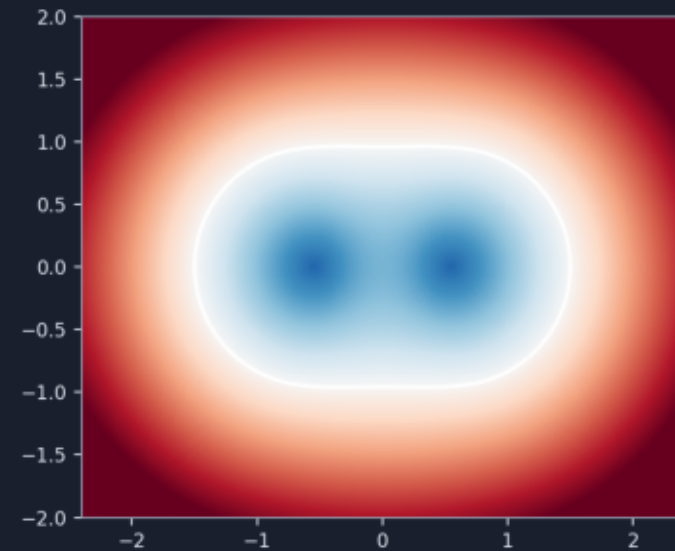
Glatte Union: $k = 0.18$



Glatte Union: $k = 0.38$

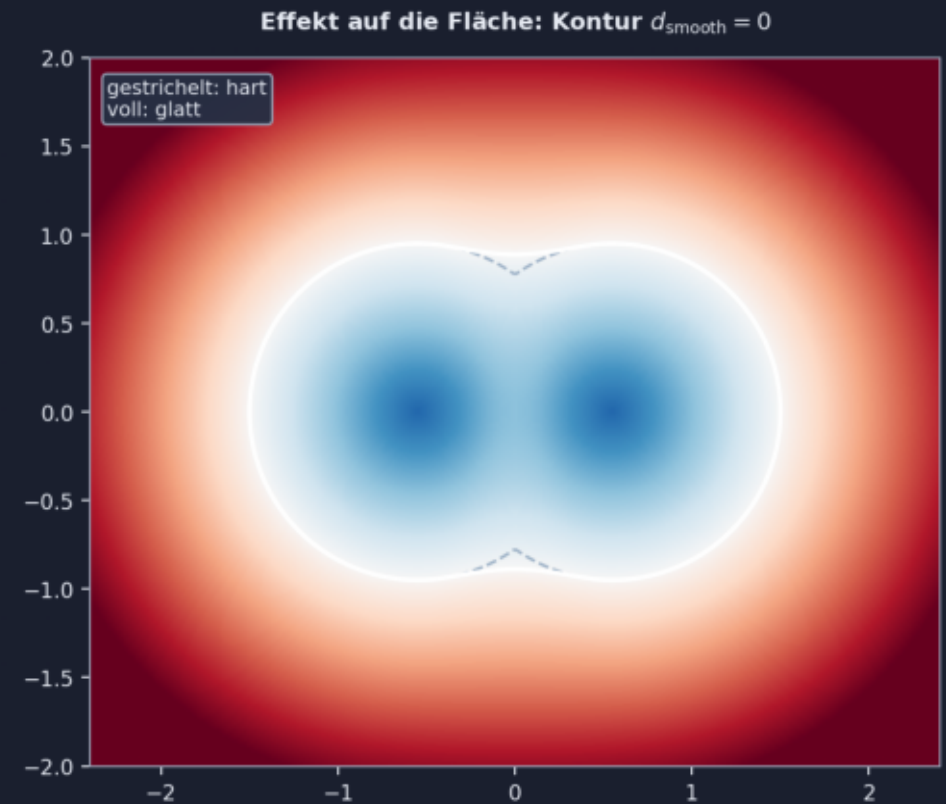
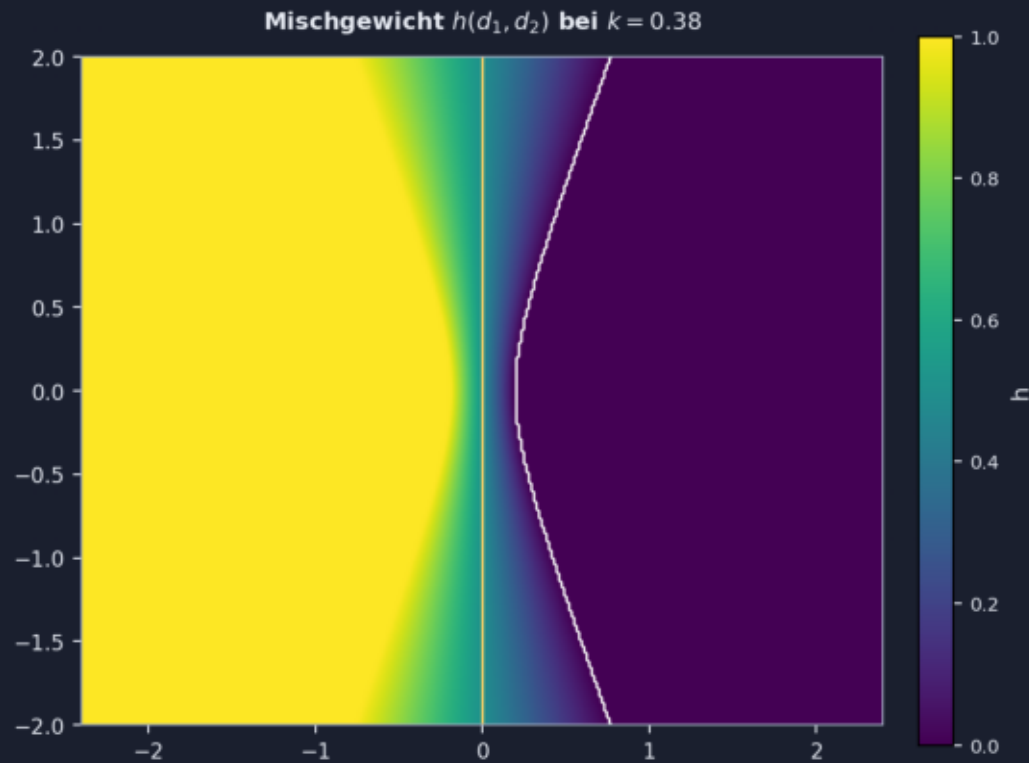


Glatte Union: $k = 0.62$

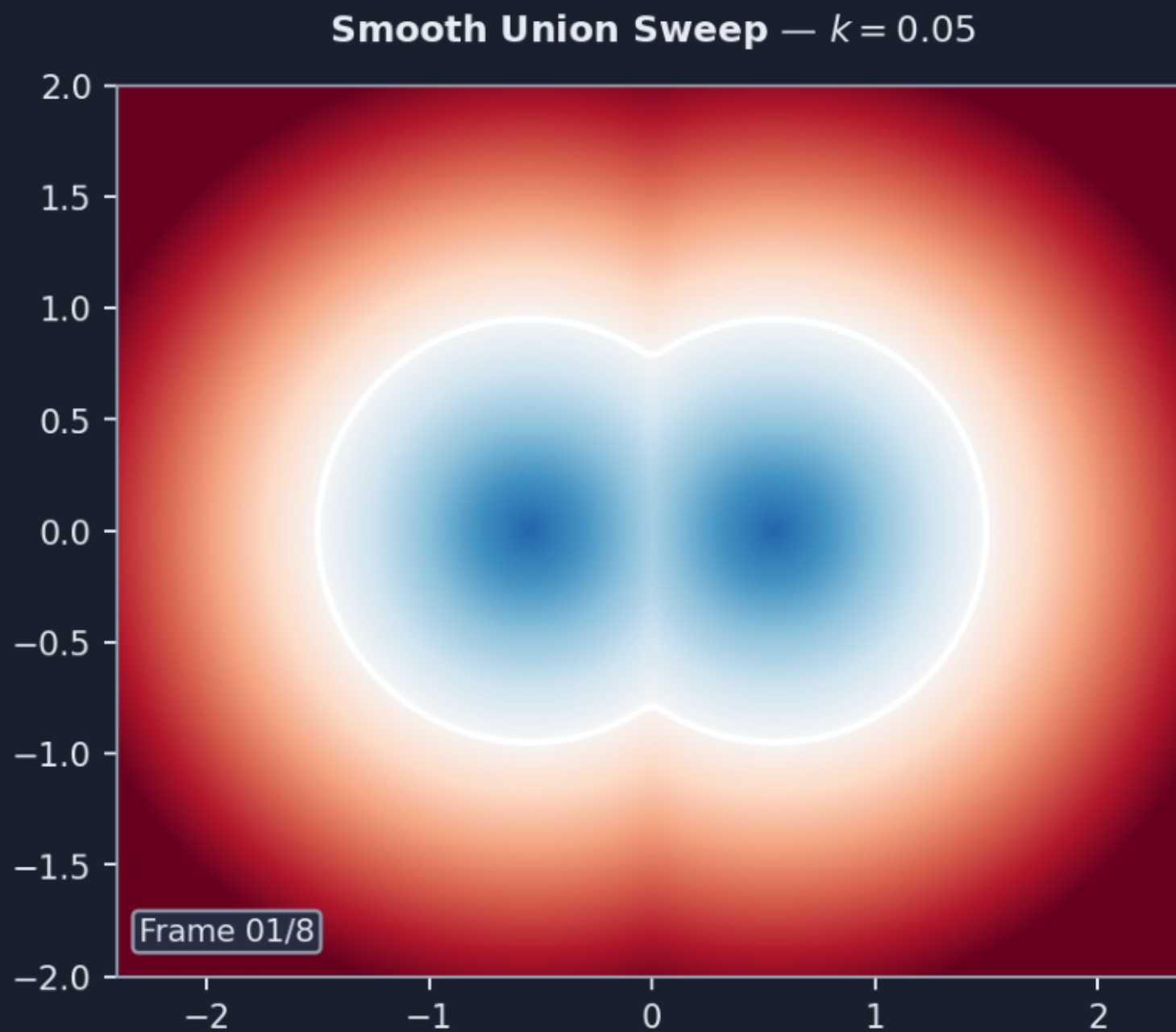


12 smooth h map

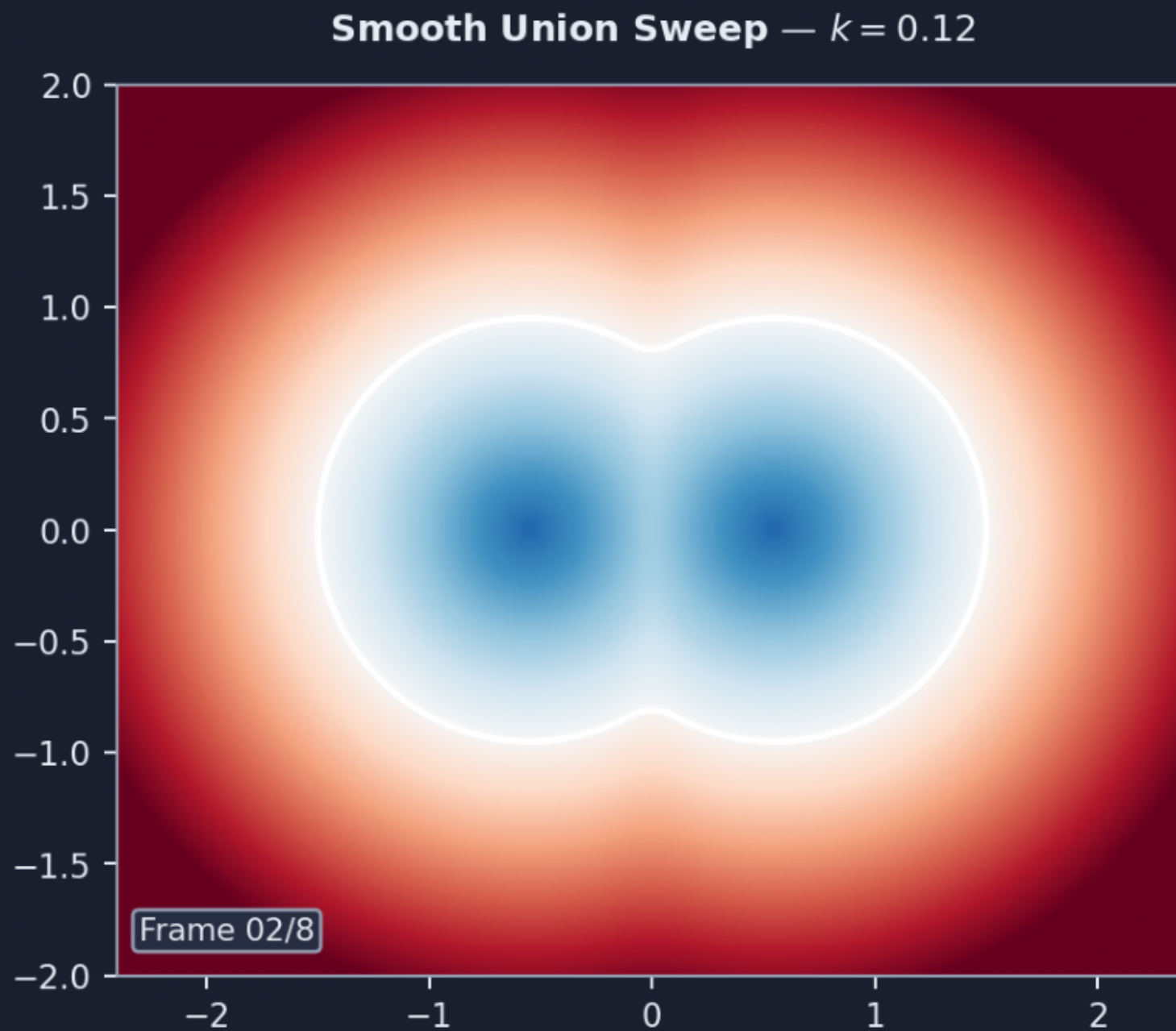
Wie smoothUnion intern arbeitet: erst h berechnen, dann distanzkorrigiert mischen



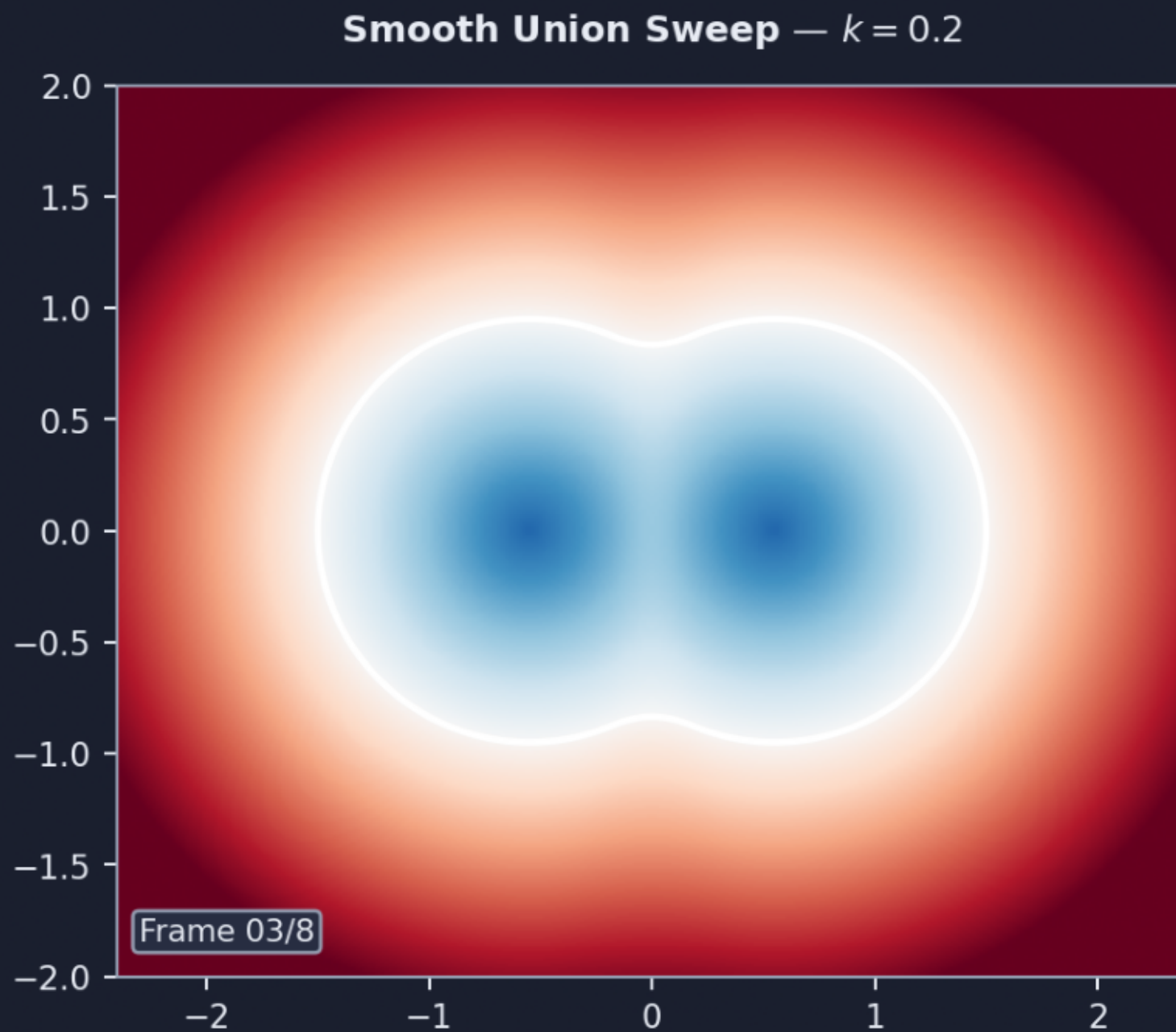
13 smooth k sweep 01



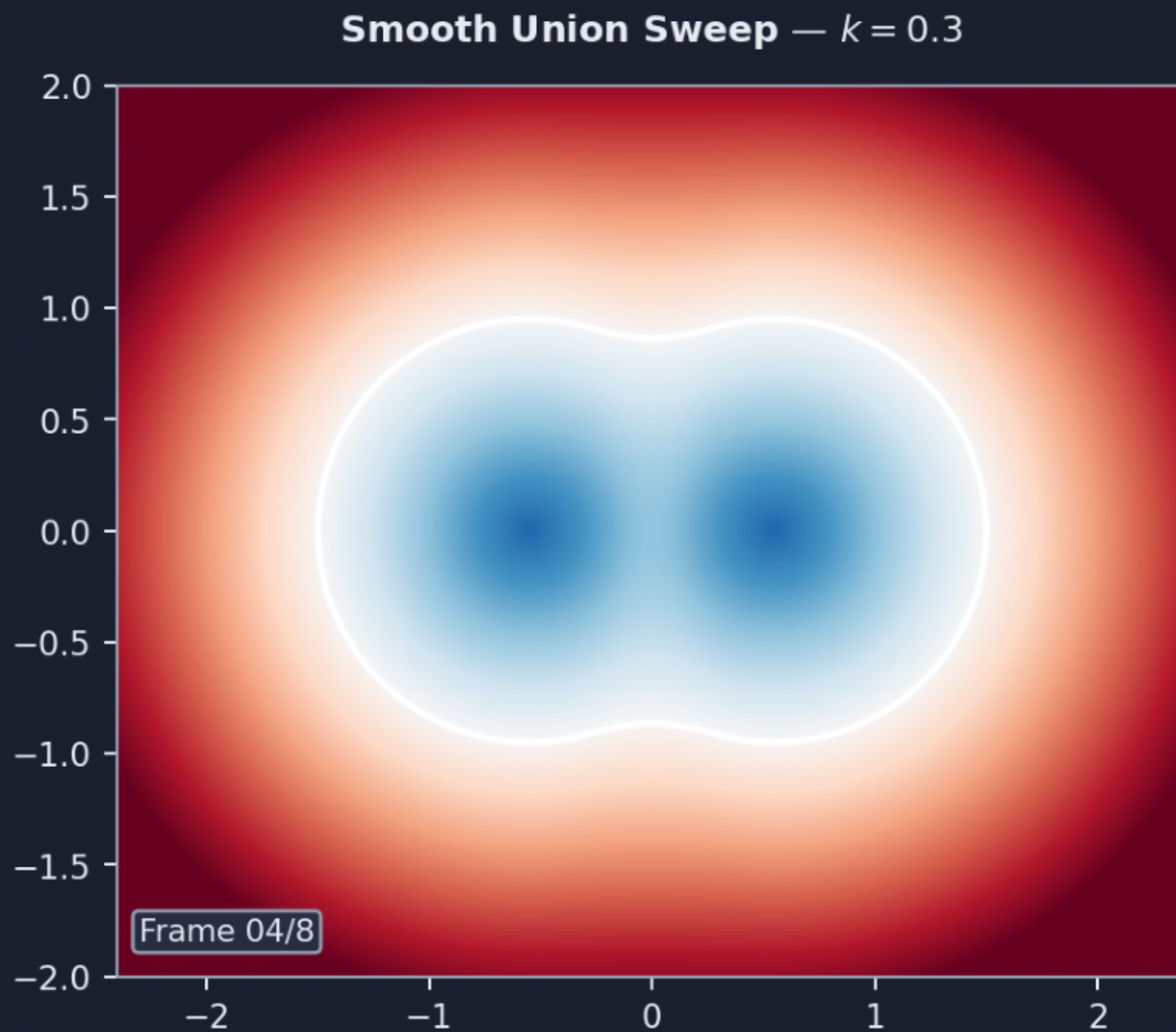
13 smooth k sweep 02



13 smooth k sweep 03

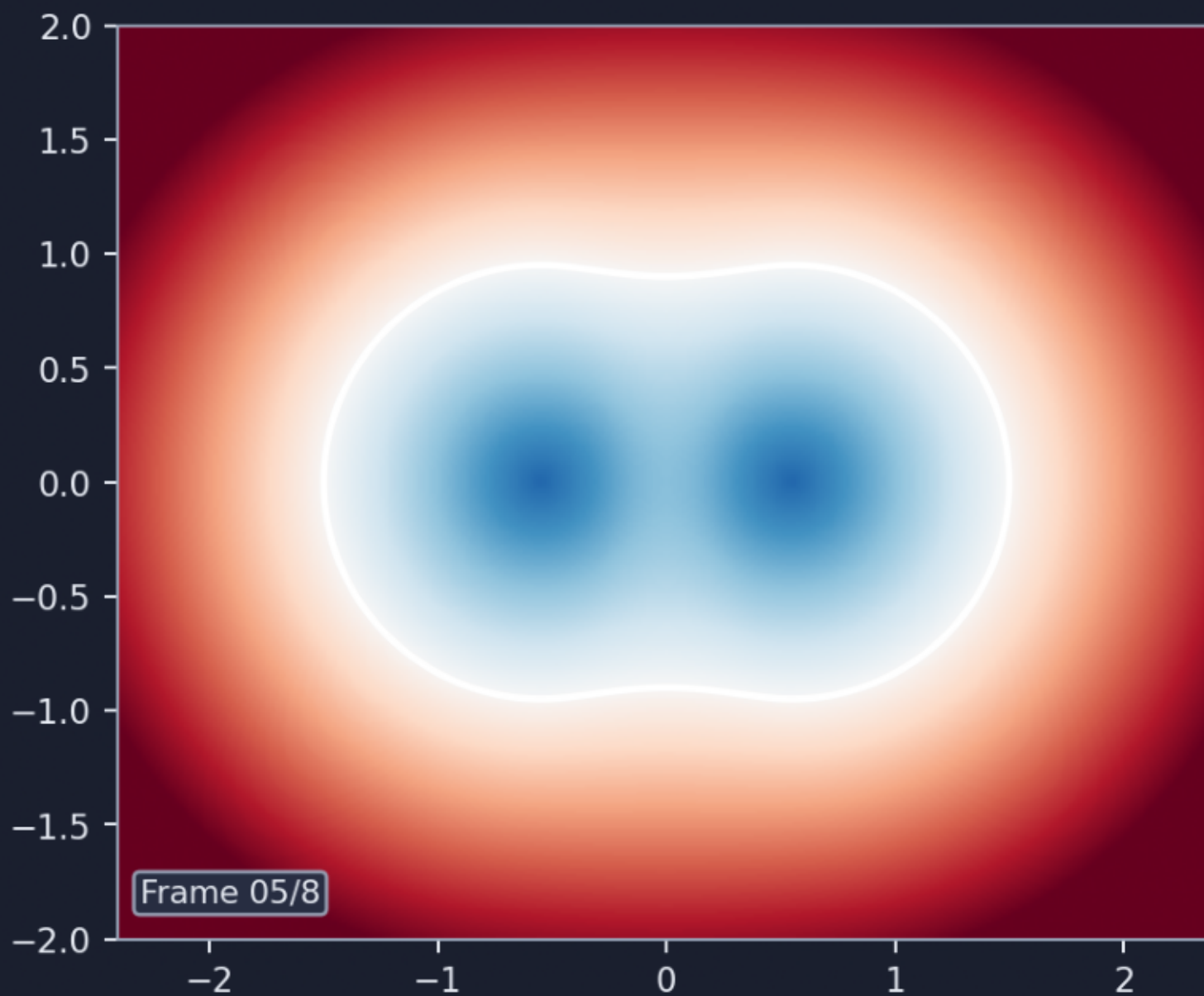


13 smooth k sweep 04

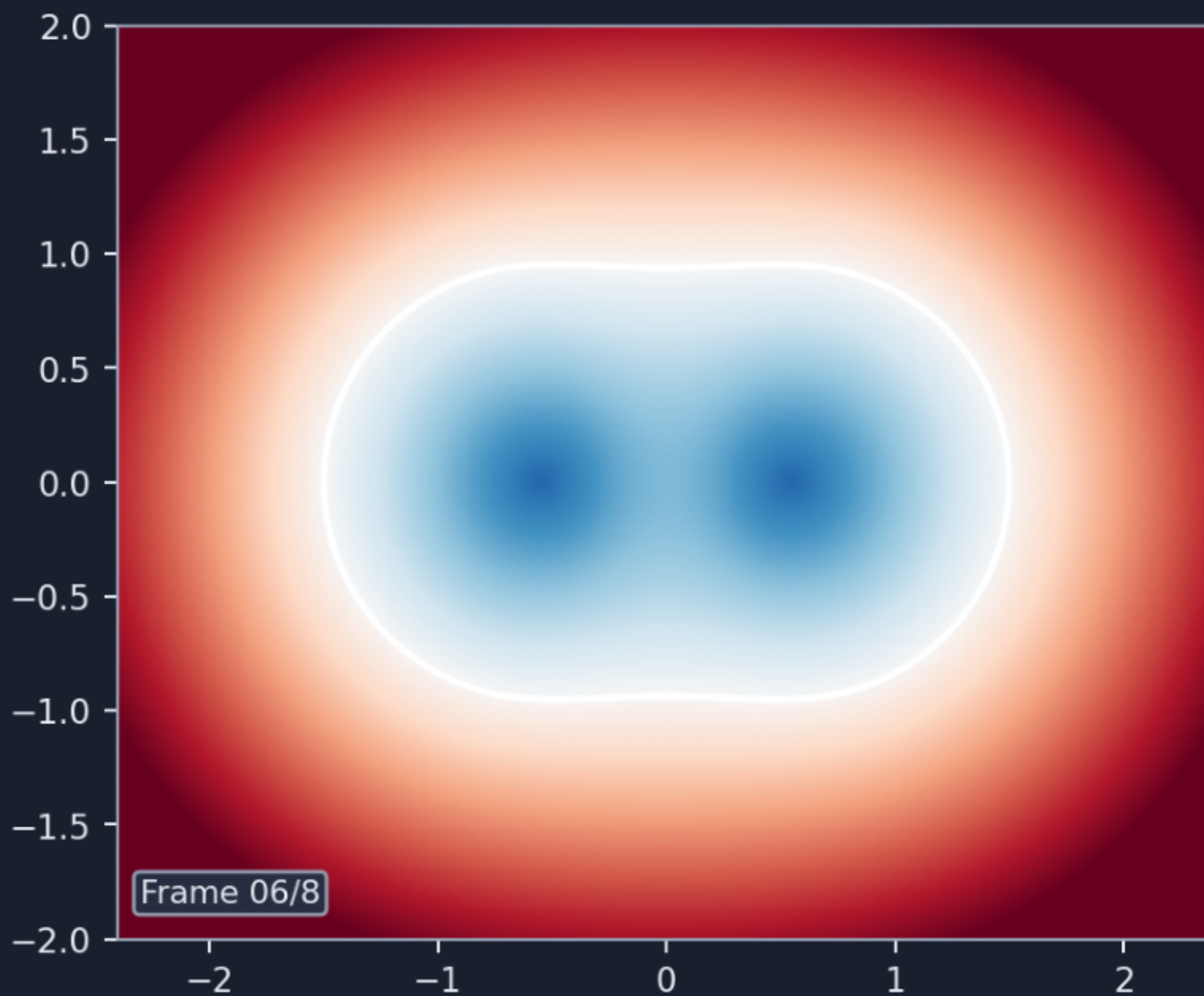


13 smooth k sweep 05

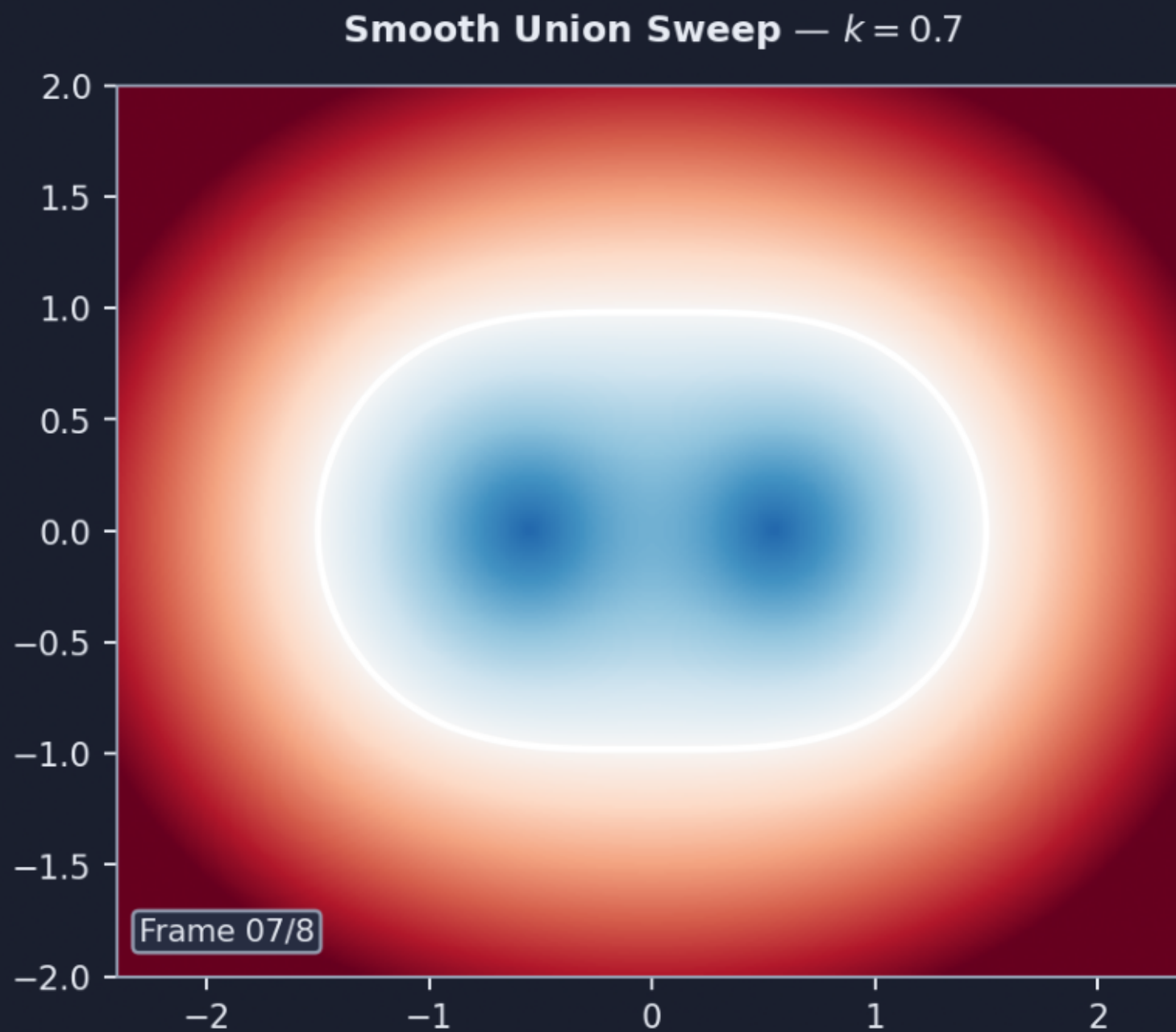
Smooth Union Sweep — $k = 0.42$



Smooth Union Sweep — $k = 0.55$

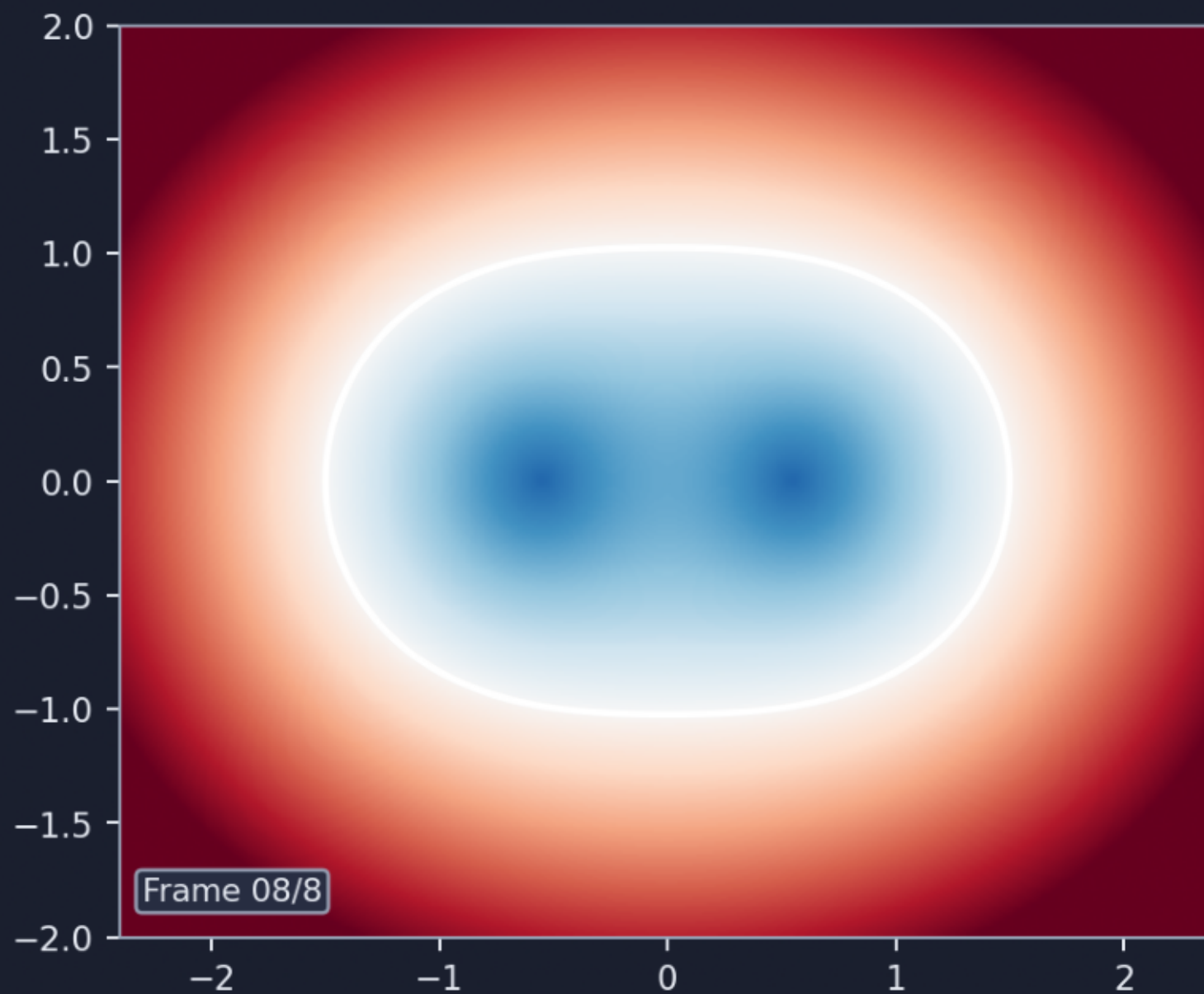


13 smooth k sweep 07



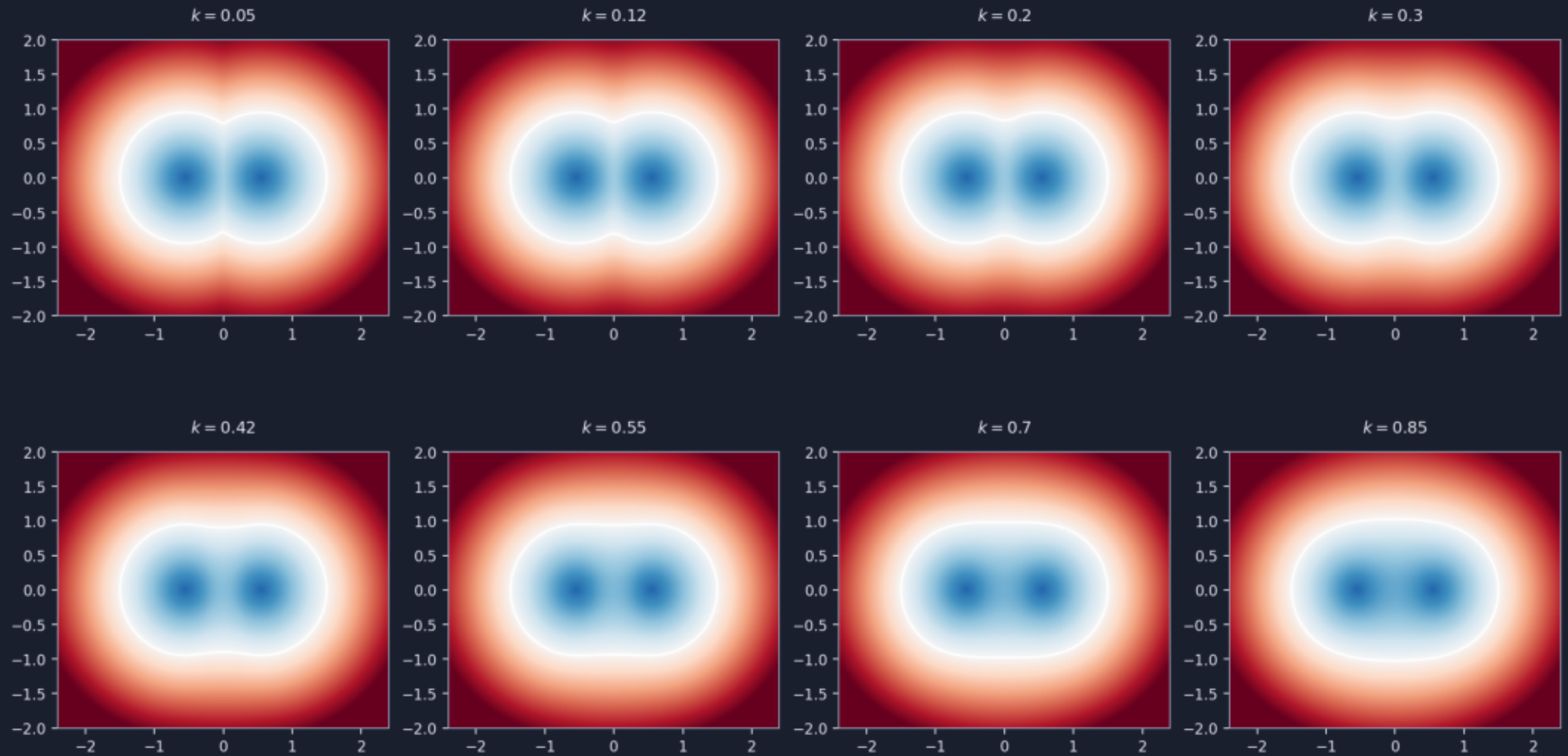
13 smooth k sweep 08

Smooth Union Sweep — $k = 0.85$



13 smooth k sweep panel

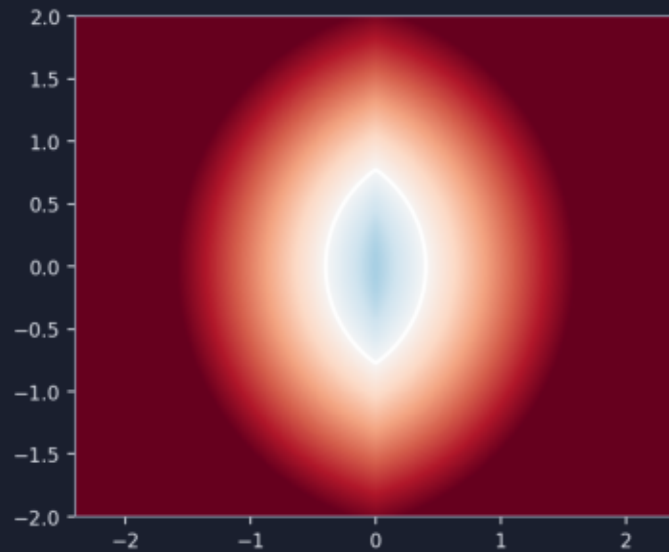
k-Sweep: kontinuierliche Verbreiterung des Blend-Bereichs



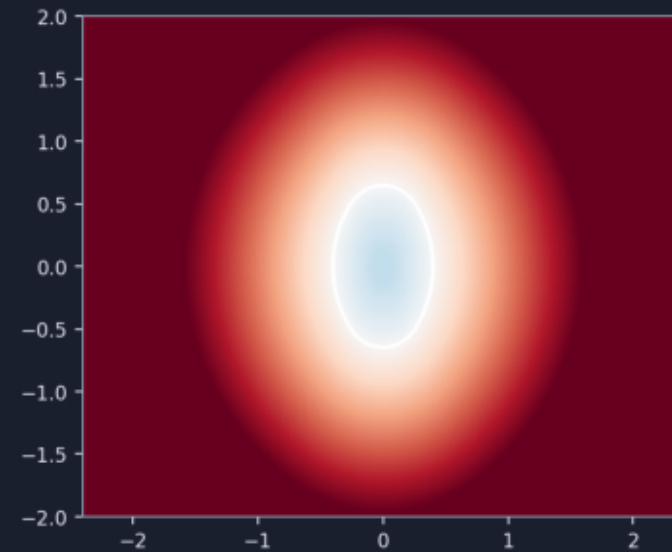
14 smooth other ops

Glatte Varianten für Schnittmenge und Differenz

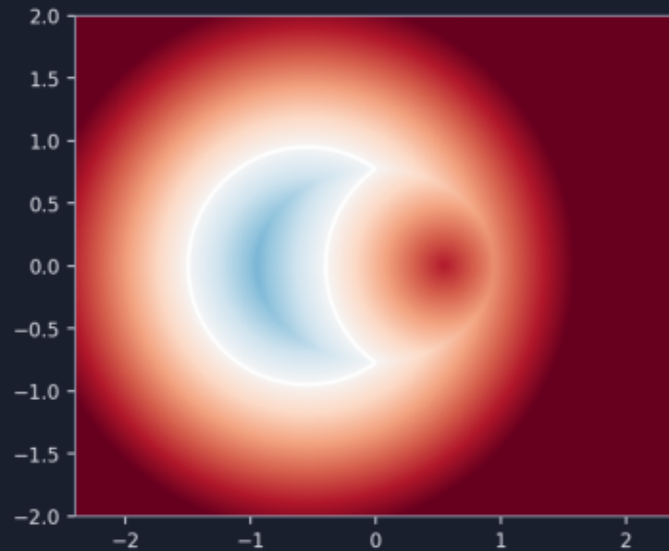
Hart: Intersection $\max(d_1, d_2)$



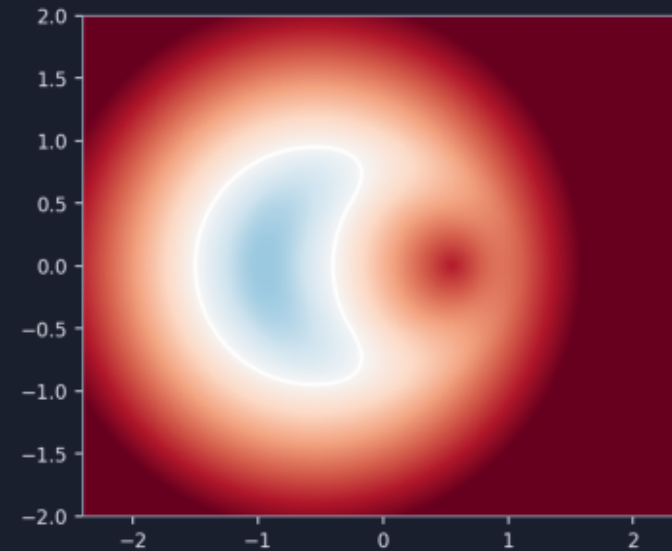
Glatt: smoothIntersection($k = 0.4$)



Hart: Difference $\max(d_1, -d_2)$



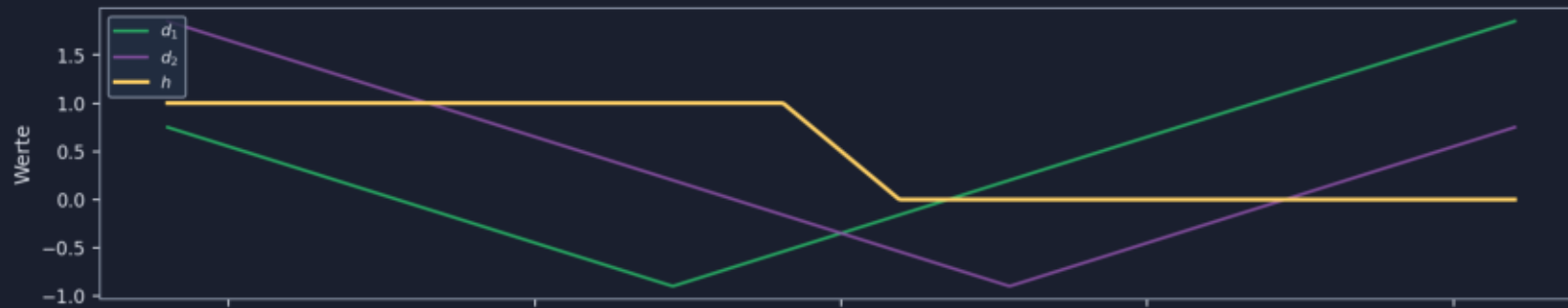
Glatt: smoothDifference($k = 0.4$)



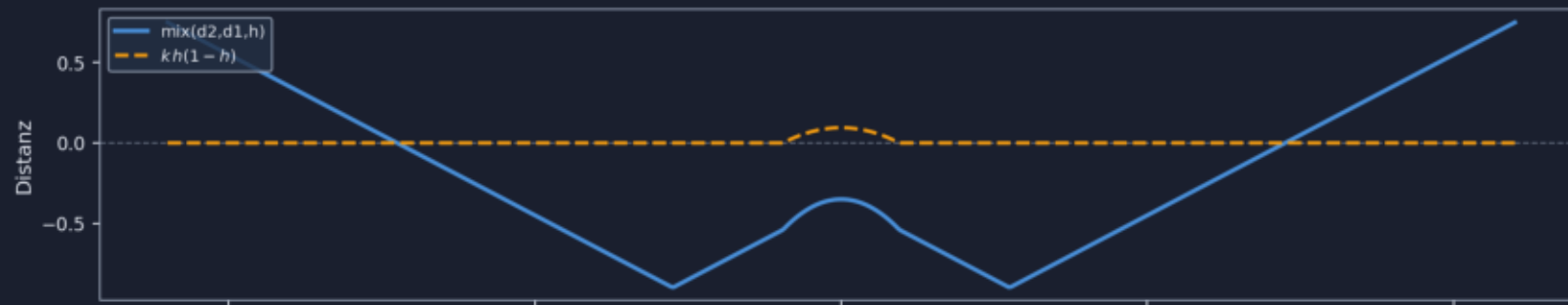
15 smooth step breakdown

Smooth-Union Schritt-für-Schritt

1) Gewicht berechnen: $h = \text{clamp}(0.5 + 0.5(d_2 - d_1)/k, 0, 1)$



2) Mischen und Korrekturterm bestimmen



3) Finale glatte Distanz: sichtbarer Übergang ohne harten Knick

